M2 joined opening: admission before recovery

15 October 2026 — analytical audit and a terminal-priority gate

Status. The user approved the three-part joined review and conditional E slowing. M2 remains selected. This note checks a shared service/allocation interface and stops at M2-A1, a possible narrow terminal-priority amendment. Neither alternative nor a complete M2 constitutive law is approved. No traffic integration, numerical gradient, calibration or optimisation occurs. All earlier mathematical sources/PDFs remain unchanged.

1. Retained physical inputs and the literal terminal rule

Retain H1's entrance histories F_j, minimum-travel eligibility P_j, finite terminal length b, jam density k_j, common advice v and known empty initial state. Write

$$P_j(t) = F_j(t - b/v), \qquad q_j = P_j - Y_j, \qquad N_j = F_j - Y_j, \quad j = E, A, B. \tag{1}$$

Cumulative mass/stock is in vehicle-equivalents; departure, demand and service rates below are in vehicle-equivalents per second. Retain

$$C(v) = \frac{wk_j v}{w + v}, \qquad 0 \le \mu(t) \le R \le C(v_-), \qquad v \in [v_-, v_+]. \tag{2}$$

μ is the one effective shared ceiling. Its M2 update is not specified by (2). Eligible lane demand has the existing interpretation

$$d_j = \begin{cases} C(v), & q_j > 0, \\ \dot{P}_j, & q_j = 0. \end{cases} \tag{3}$$

The inherited strict allocator is

$$g_A = \min\{d_A, \mu\}, \qquad g_E = \min\{d_E, [\mu - g_A]_+\}, \qquad \dot{Y}_j = g_j. \tag{4}$$

Strict-priority consequence. Whenever $q_A > 0$, (2)–(3) give $d_A = C(v) \ge \mu$, hence

$$q_A > 0 \quad \Longrightarrow \quad g_A = \mu, \quad g_E = 0. \tag{5}$$

The consequence holds for every admissible service history and every recovery memory. Lowering μ cannot by itself create an E allocation. This is temporary exclusion during A backlog, not proof of permanent starvation of a finite cohort. It does not rule out E entry while A is unqueued or A delay caused by earlier E entry.

Interpretation limit. Equations (3)–(5) average within-lane service; they do not identify physical gaps. Giving E passage during $q_A > 0$ by a new demand adapter would change this contract just as changing (4) would. It is not a local transcription correction.

2. A concrete alternative allocation, not a recovery law

The merge template in Jin, Gan and Lebacque, section 6, equations (15) separates downstream service and priority. Reuse only its allocation shape, with a proposed E priority weight $0 \le \beta < 1$:

$$\begin{aligned} g_E &= \min\{d_E, \max\{\mu - d_A, \beta\mu\}\}, \\ g_A &= \min\{d_A, \max\{\mu - d_E, (1-\beta)\mu\}\}. \end{aligned} \tag{6}$$

This is a proposed M2 adaptation of that allocation; the source's phenomenological capacity-drop law is not imported. A weight favoring A would use $\beta < 1/2$, but no value or calibration is selected. The weight is a shared physical/model parameter, not another control or an observer privilege. At $\beta = 0$, (6) reproduces (4).

For all nonnegative demands and service,

$$0 \le g_j \le d_j, \qquad g_A + g_E = \min\{d_A + d_E, \mu\}. \tag{7}$$

To verify, first take $d_A + d_E \le \mu$: both minima equal their own demands. Otherwise, if $d_E \le \beta\mu$, E gets its demand and A the remainder; if $d_A \le (1-\beta)\mu$, reverse those roles. In the remaining case each receives its weighted share. These cases exhaust the domain, including equalities and $\mu = 0$.

If both queues are positive, $d_A = d_E = C(v) \ge \mu$, so

$$(g_A, g_E) = ((1-\beta)\mu, \beta\mu). \tag{8}$$

If only A is queued, $g_E = \min\{d_E, \beta\mu\}$; an E stream with positive demand can therefore enter before the A queue empties. Unused E allocation returns to A. A lone stream can use the whole ceiling. Thus β does not impose a fixed completed merging proportion outside contention.

Hand case, not a rollout. At $\mu = 600$ vehicle-equivalents/hour and saturated eligible A/E demands, strict priority gives $(600, 0)$; illustrative $\beta = 1/4$ gives $(450, 150)$. These are exact allocations, not proposed traffic parameters or empirical throughput. The common total is unchanged. If total demand is 400, (7) passes 400 under either rule; no artificial queue or slowing is implied by (6).

Scope. Changing (4) to (6) with $\beta > 0$ changes terminal right-of-way/service allocation. It leaves H1's own-lane-first lateral rule, cutoff, storage, advice and local completion untouched. The allocation alone specifies neither an acceptable physical gap, crossing speed, maneuver friction nor recovery. It is not a complete M2 law.

3. Kinematic and recovery guardrails

For a triangular fundamental diagram, a congested branch compatible with a rate $0 < f < C(v)$ is

$$k_{\mathrm{c}}(f) = k_j - f/w, \qquad u_{\mathrm{c}}(f) = \frac{wf}{wk_j - f}, \qquad C(u_{\mathrm{c}}(f)) = f. \tag{9}$$

The free branch at the same rate has speed v. A queue/regime can motivate examining (9); the lateral insertion trajectory does not follow uniquely from a point count. No actual speed is assigned to absent traffic.

The source Leclercq et al., section 3.2 uses the congested-entry regime for its flow/speed relation. It does not prove a universal empty-start point-entry map. Its mean-capacity process and our finite transient account remain distinct.

Why an automatic A feedback needs care. Suppose a tentative rule uses (9) for queued A and takes the fresh isolated loss amplitude to be $D_A(0) = R - C(u_A)$. Under strict priority, (5) gives

$$q_A > 0, \quad u_A = u_{\mathrm{c}}(g_A), \quad g_A = \mu \quad \Longrightarrow \quad D_A(0) = R - \mu. \tag{10}$$

The supposed new A-loss amplitude is exactly the shared service reduction already acting. Adding a fresh independent recovery copy for every such follower needs another physical justification; observing low A speed alone does not supply it. Under (8) it would even include the part of service allocated to E. This diagnostic does not rule out a separately justified A startup/acceleration mechanism or prove every possible A model duplicates loss. No such extra source is selected.

Shared-account contract. Let Z denote sufficient predicted aggregate state/history, not traffic mass, and let a future qualified law produce $\mu = \mathcal{M}(Z, \text{current state})$. Admissions can update Z using their amount/type and justified kinematics. A single output μ is compatible with richer internal memory. Neither an unproved one-state reduction nor independent constraints per past entry are assumed. An E passage with no actual disturbance must not acquire a fixed slow-entry penalty merely because it is E.

Still open. Conditional admission/opportunity, actual departure speed, recovery overlap and causal closure remain unformed. Neither (6) nor a larger memory automatically supplies these. Initial ordinary underload may still be a benign regime; a compulsory interaction must be specified before claiming the model generates a consequential episode. No exclusive slowing-origin choice is reinstated.

4. Joined interfaces that can already be checked conditionally

Assume departures are conservative, absolutely continuous, minimum-travel eligible and demand/service feasible. The fixed finite-terminal domain

$$H = k_j b > M, \qquad 0 \le \dot{F}_j \le C(v), \qquad F_j \le M \tag{11}$$

protects H1's handoff under either allocation. Before b/w, vacancy is at least $(k_j w - C(v))t > 0$; afterwards,

$$V_j(t) = Y_j(t - b/w) + H - F_j(t) \ge H - M > 0. \tag{12}$$

Physical stock remains $N_j = F_j - Y_j$; Z is never added to stock. These are sufficient interface bounds, not a general upstream spillback solver.

For the uniform initial footprint $[0, \ell]$ on the road of length L, complete local cohort time has the unchanged conservative identity

$$J(v) = \frac{M(L - \ell/2)}{v} + \int_0^\infty (q_A + q_E + q_B)\, dt. \tag{13}$$

It follows by writing $M - \sum_j Y_j = (M - \sum_j P_j) + \sum_j q_j$ and integrating each travel-eligibility delay. A and E queue areas both count. Post-crossing recovery may affect following departures but contributes no second post-crossing vehicle-time to the local objective.

A sufficient clearance criterion, not a new M2 proof. Set $Q = q_A + q_E$ and $t_* = L/v$, after which all traffic is eligible and no new external traffic arrives. B retains its independent admitted domain and $Y_B = P_B$, so B is clear by t_*. If a candidate guarantees a finite total loss budget

$$0 \le \mu \le R, \qquad \int_{t_*}^\infty (R - \mu(t))\, dt \le \Lambda < \infty, \tag{14}$$

then work conservation (7), or (4), gives $\dot{Q} = -\mu$ whenever $Q > 0$. If it had not cleared by $T > t_*$,

$$Q(T) = Q(t_*) - R(T - t_*) + \int_{t_*}^T (R - \mu(t))dt \le Q(t_*) - R(T - t_*) + \Lambda. \tag{15}$$

Thus clearance occurs by $t_* + [Q(t_*) + \Lambda]/R$. The argument permits temporary zero service and assumes no artificial positive floor. M2 must establish (14) or another valid clearance argument for its exact law; M1's finite-kernel proof does not transfer automatically.

Limits and gate. Neither a complete coupled-law uniqueness proof nor a reachable consequential MP6 domain is supplied. Known initial state plus a uniquely autonomous law would permit the familiar ending-statistic initialization, but that conditional statement does not establish decision relevance or a useful gradient. M2-A1 asks whether to retain (4) or reopen terminal allocation for positive-weight (6). Both preserve conditional E slowing and the model–SUMO S-P–SUMO S-L ladder. Freeze no new priority, demand domain, information channel or recovery law at this audit.